Tetris Trials

Lars van de Corput & Gijs Bolwerk

Tetris trials is a different twist on Tetris, mixing it with objectives and tower defence. The goal of the game is to rack up as much score as you can, while pushing ghosts back by completing objectives and getting buffs. Failing to complete these objectives can have negative effects, so you should be cautious. As the game goes on, it will become more challenging. Aim for the highest score!

https://github.com/Larscorp/CBL-Project2

Advanced topics:

Git branching workflows

Modern Java Features (Lambda Expressions for objectives)

#	Name	How to demo	Notes
1	Main Menu	Start the game; the main menu with options to start a new game and quit show up.	Starting point of the game.
2	Display game board	Click on start on the main menu; empty game board shows up.	Board is 10x22 cells with 2 invisible cells on top.
3	Play	Click on play; the board is reset and a new piece appears at the top.	Should reset all active elements such as score, objectives, board etc.
4	Spawn pieces	Start a game; a piece should spawn at the top of the board.	Pieces spawn in a starting position.
5	Piece movement	Click the left and right arrow keys; The current piece moves left and right respectively.	Restricted by boundaries.
6	Automatic falling	Start a game; the current piece starts falling automatically.	Based on level system.

7	Collision detection	Move a piece against the wall; The piece stops and can't move past the wall. Drop a current piece down on another piece; The piece locks and a new piece spawns on top.	Prevents overlapping and going out of bounds.
8	Piece rotation	Press the up arrow key; The piece rotates clockwise.	Clockwise and counterclockwise implementation.
9	Line clearing	A line is full; The line is cleared and every line above is shifted down.	The completed line is removed.
10	Game over	The board is filled in a way that a next piece can't spawn or hp is 0; The game shows a game over screen.	Game over screen has final score, quit, go to main menu and retry options.
11	Bag	Start the game and drop 14 different pieces; Every piece can only occur 2 times, then it is different from the last piece.	14-bag system.
12	Hold	Press c; The current piece gets put in the hold slot. The piece that was in the hold slot becomes the current piece.	You cant hold more than once per current piece.
13	Score	Clear a line; The score goes up, based the formula (base score * level)	Points are awarded for clearing lines, multiplied by the current level.
14	Main menu	Click return to main menu; return to the main menu.	Stops gameplay

5	Quit	Click quit; the game closes.	Closes the game window.
16	Objective	Start a game; The objective ui displays at the top of the screen, displaying an objective.	Timed objective.
17	Objective complete	Complete an objective before the timer expires; A ghost is removed and a new objective is given.	No ghosts left and objective complete means getting a buff*
18	Ghosts	Play the game; after decreasing amount of seconds, ghosts spawn at the top.	Higher level means faster ghost speed and spawns.
18	Objective failure	Let the timer expire before completing an objective; A debuff** is applied and a new objective is given.	Doesn't do anything to ghosts.
19	Objective buff*	Complete an objective; A positive effect is applied.	Picked from a random pool.
20	Objective debuff**	Fail an objective; A negative effect is applied	Picked from a random pool.
21	Scoreboard	Open the main menu; A screen shows the highest achieved scores.	Top 5 highest scores.
22	Save game state	Close and reopen the game; The high scores are saved on the scoreboard.	Stores in scores.txt
23	Increasing difficulty	Clear 10 lines; The level indicator goes up by one and the speed at which the game plays increases.	Based on a formula.